

NETWORKS AND COMPLEXITY

Solution 14-1

*This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 14.1: Push and Pull [Lvl. 1]

Consider a steady state with the abstract Jacobian

$$\mathbf{J} = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix}$$

Compute the eigenvalues of this matrix to show: a) The off-diagonal elements only in the form of the weight of the cycle $w = J_{12}J_{21}$ matters for the eigenvalues, b) if $w = 0$ the eigenvalues are J_{11} and J_{22} . c) if $w < 0$ the eigenvalues are pulled together, d) if $w > 0$ the eigenvalues are pushed apart.

Solution

Part (a). We have to solve

$$\begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix} \quad (1)$$

The first line reads

$$J_{11}x + J_{12}y = \lambda x \quad (2)$$

and hence

$$\lambda = J_{11} + J_{12} \frac{y}{x} \quad (3)$$

similarly the second line gives us

$$J_{21}x + J_{22}y = \lambda y. \quad (4)$$

Here we don't care about the eigenvector. We just want to discover how the eigenvalues depend on J_{11} , J_{12} , J_{21} , and J_{22} . Hence we seek to eliminate x and y from the equation. Since the first line gave us a condition that only contains the fraction y/x let's see if we can create this fraction again. So, we divide the condition from the second line by x which yields

$$J_{21} + J_{22} \frac{y}{x} = \lambda \frac{y}{x} \quad (5)$$

and hence

$$\frac{J_{21}}{\lambda - J_{22}} = \frac{y}{x} \quad (6)$$

which we can substitute into the first condition to find

$$\lambda = J_{11} + \frac{J_{12}J_{21}}{\lambda - J_{22}} \quad (7)$$

Part (b). At this point we have already reached our first goal (a) as we can now see that J_{12} and J_{21} only appear in form of the product $J_{12}J_{21}$ which is our cycle weight w . Of course, the condition is actually our characteristic polynomial which we can now write in the standard form

$$\lambda^2 - J_{22}\lambda - J_{11}\lambda + J_{11}J_{22} - w = 0 \quad (8)$$

We can now just factorize the characteristic polynomial

$$(\lambda - (J_{22} + J_{11})/2)^2 - (J_{22} + J_{11})^2/4 + J_{11}J_{22} - w = 0 \quad (9)$$

$$(\lambda - (J_{22} + J_{11})/2)^2 = J_{22}^2/4 + J_{11}^2/4 - J_{11}J_{22}/2 + w \quad (10)$$

$$\lambda - (J_{22} + J_{11})/2 = \pm \sqrt{(J_{22} - J_{11})^2/4 + w} \quad (11)$$

$$\lambda = (J_{22} + J_{11})/2 \pm \sqrt{(J_{22} - J_{11})^2/4 + w} \quad (12)$$

$$(13)$$

We have now almost reached the next goal. If $w = 0$ then we can further simplify

$$\lambda = \frac{J_{22} + J_{11}}{2} \pm \sqrt{\frac{(J_{22} - J_{11})^2}{4}} = \frac{J_{22} + J_{11}}{2} \pm \frac{J_{22} - J_{11}}{2} \quad (14)$$

So we find the eigenvalues

$$\lambda_1 = \frac{J_{22} + J_{11}}{2} + \frac{J_{22} - J_{11}}{2} = J_{22} \quad (15)$$

$$\lambda_2 = \frac{J_{22} + J_{11}}{2} - \frac{J_{22} - J_{11}}{2} = J_{11} \quad (16)$$

$$(17)$$

Which shows (b): if the weight is zero, the eigenvalues are J_{11} and J_{22} .

Part (c) and (d). Now let's return to the case where the weight is not zero. From the equation

$$\lambda = \frac{J_{22} + J_{11}}{2} \pm \sqrt{\frac{(J_{22} - J_{11})^2}{4} + w} \quad (18)$$

we can already see that the separation between the two eigenvalues grows when $w > 0$ and shrinks when $w < 0$ which covers points (c) and (d). But can we actually see how strong the pull or push from the cycle is?

Ideally, I want an equation such as $\lambda_1 = J_{22} + x$, where x is the effect of the push or pull from the cycle. So, let's substitute this in. Because we focused on the eigenvalue λ_1 , we only need to consider the positive case, so

$$J_{22} + x = \frac{J_{22} + J_{11}}{2} + \sqrt{\frac{(J_{22} - J_{11})^2}{4} + w} \quad (19)$$

which we can also write as

$$x = -\frac{J_{22} - J_{11}}{2} + \sqrt{\frac{(J_{22} - J_{11})^2}{4} + w} \quad (20)$$

It is interesting that J_{11} and J_{22} now only appear in expressions of the form $(J_{22} - J_{11})/2$ which is half the distance between J_{11} and J_{22} or in other words the distance between J_{22} and the midpoint between J_{11} and J_{22} . Let's call this h . We define

$$h = \frac{J_{11} - J_{22}}{2} \quad (21)$$

Substituting simplifies our condition for x to

$$x = -h + \sqrt{h^2 + w} \tag{22}$$

This is about as nice as we can make it. From this form the results (b,c,d) are quite obvious.

Equivalently we can say that the total distance the eigenvalue is from the halfway point between the eigenvalues is

$$x + h = \sqrt{h^2 + w}. \tag{23}$$